

	Course	Databases and Information Systems			2026
	Exercise Sheet	3			
	Points	–			
	Release Date	June 23 rd 2026	Due Date	June 29 th 2026	

Exercise Sheet 3: Forecasting

With this exercise sheet, you will use the data from the previous exercises to implement a forecasting algorithm to present at the end of the semester. This requires working practically. Unfortunately, our exercise rooms this year are not equipped with PCs and the PC pools are not entirely usable, yet, so please bring a laptop or tablet if you want to work in the exercise rooms or need to show something to explain a question. If you do not have any portable device you can use for the exercise, please consider using the lending service of the General Students' Committee¹.

Please read the whole exercise sheet before starting to work. The results are not graded as they are highly individual to the setup and method you are using. Please continue working in the same groups as in the previous exercise sheets. You are free to use a programming language of your choice, but be aware that there might be no support if you chose an exotic language.

1 The task

Forecast the **daily** usage of ingredients in a café using any **non-ML** forecasting technique of your choice that was covered in the lecture. Use the dataset from the first exercise and your completed dataset from the second exercise for this task. Then, create the following images:

- A plot showing the original data and the forecasted data for days not included in the original dataset
- A plot showing the original data and the forecasted data for days included in the dataset (for evaluation)
- An image showing the architecture you built to solve the task, e.g. including the data, databases and data flows involved. Some inspiration can be found in the demo papers from this year's SIGMOD conference, e.g.,
 - <https://dl.acm.org/doi/pdf/10.1145/3788853.3801615> (Fig. 3)
 - <https://dl.acm.org/doi/pdf/10.1145/3788853.3801592> (Fig. 1)
 - <https://dl.acm.org/doi/pdf/10.1145/3788853.3801578> (Fig. 1)
 - <https://dl.acm.org/doi/pdf/10.1145/3788853.3801586> (Fig. 2)

Include where your forecasting is executed, e.g., as a UDF in the database system or in an external script after aggregating and extracting the relevant values.

Further, if a recipe provides a choice between two ingredients, define which ingredient you assume in these cases. The estimation of the model's parameters might be done naively if this leads to sufficient results.

¹<https://www.asta-uhh.de/4-service/1-verleihservice.html>, accessed 15/04/2026

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2 Requirements

Your demo must fulfill a number of requirements:

- The forecasting algorithm must be implemented by yourself, i.e., do not use any existing forecasting tools
- The data must be drawn from at least one database system, not from the original files
- There must be a user interface. A command line interface is sufficient, a graphical user interface is nice but not required
- At least one parameter of the forecasting model must be adjustable by the user
- The user should be able to switch between the forecast of at least two ingredients, e.g. milk and sugar.
- The forecast should finish within a reasonable amount of time, i.e., **not** take several minutes or even hours
- The plot must be generated automatically with little or no user interaction
- Use a part of the original data to validate your forecasting results, i.e., forecast a part of the series that already has entries to compare the result with

3 Upload

Go to moodle and navigate to the task 'Upload for Exercise Sheet 3'². Upload the following files:

- A plot with the forecast for the future. Include the chosen parameter(s).
- A plot with the forecast used for evaluation. Include the chosen parameter(s).
- The architecture figure
- A plain text containing the name of the used forecasting algorithm, the parameters the user can change, and the name of the forecasted ingredients
- A file containing only the source code for the forecasting, no other code, e.g., not the connection to the database, any data exports, or the plotting script

Again, it is sufficient if one member of a group uploads the results.

²<https://lernen.min.uni-hamburg.de/mod/assign/view.php?id=284187>